



CHANDIGARH UNIVERSITY
APEX INSTITUTE OF TECHNOLOGY



ASSIGNMENT 1

Subject: Computer Networks

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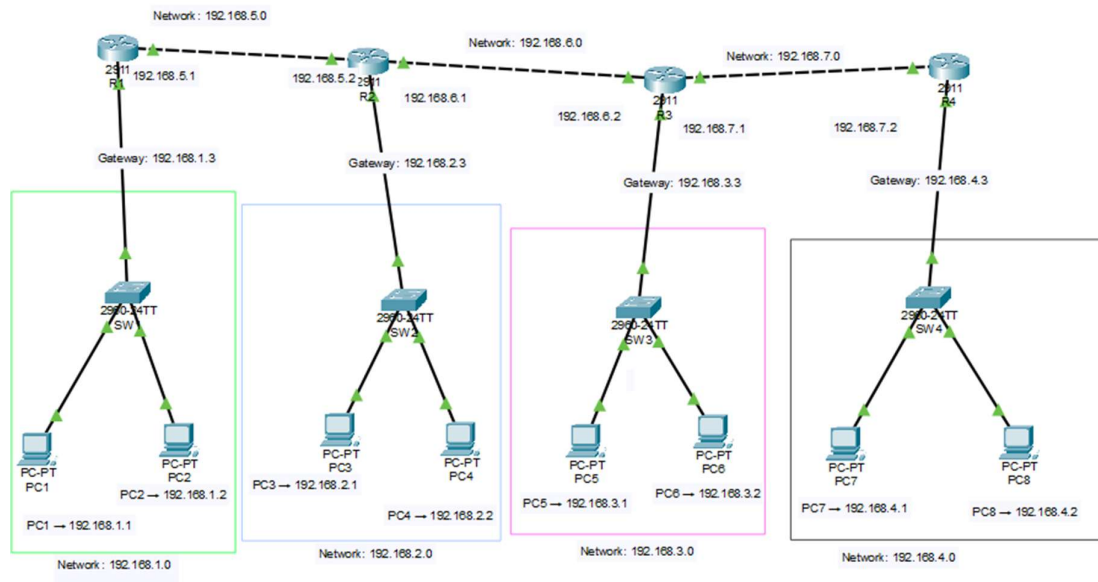
Submitted To:

Mr. Alok Singh Rathore

Q1. Develop Static Routing with 4 PC's, 4 Switches & 8 PCs. Make sure to highlight the Networks with their network ID's. IP Addresses should be Written in text format as well.

Answer:

In this topology, 4 routers, 4 switches, and 8 PCs are connected to form multiple networks. Static routes are configured on each router so that PCs from different networks can communicate with each other.



1. Network Addressing

The topology consists of 7 different networks, using the subnet mask:

Subnet Mask: 255.255.255.0 (/24)

Network	Network ID	Devices / Connection
LAN 1	192.168.1.0/24	R1 – SW1 – PC1, PC2
LAN 2	192.168.2.0/24	R2 – SW2 – PC3, PC4
LAN 3	192.168.3.0/24	R3 – SW3 – PC5, PC6
LAN 4	192.168.4.0/24	R4 – SW4 – PC7, PC8
Network 5	192.168.5.0/24	R1 ↔ R2

Network	Network ID	Devices / Connection
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Network 6	192.168.6.0/24	R2 ↔ R3
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Network 7	192.168.7.0/24	R3 ↔ R4
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2. IP Addressing of PCs

PC	IP Address	Subnet Mask	Default Gateway
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PC1	192.168.1.1	255.255.255.0	192.168.1.3
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PC2	192.168.1.2	255.255.255.0	192.168.1.3
-----	-------------	---------------	-------------

PC3	192.168.2.1	255.255.255.0	192.168.2.3
-----	-------------	---------------	-------------

PC4	192.168.2.2	255.255.255.0	192.168.2.3
-----	-------------	---------------	-------------

PC5	192.168.3.1	255.255.255.0	192.168.3.3
-----	-------------	---------------	-------------

PC6	192.168.3.2	255.255.255.0	192.168.3.3
-----	-------------	---------------	-------------

PC7	192.168.4.1	255.255.255.0	192.168.4.3
-----	-------------	---------------	-------------

PC8	192.168.4.2	255.255.255.0	192.168.4.3
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3. IP Addressing of Routers

Router	Interface	IP Address	Connected Network
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R1	G0/0	192.168.1.3	192.168.1.0/24
----	------	-------------	----------------

R1	G0/1	192.168.5.1	192.168.5.0/24
----	------	-------------	----------------

R2	G0/0	192.168.2.3	192.168.2.0/24
----	------	-------------	----------------

R2	G0/1	192.168.5.2	192.168.5.0/24
----	------	-------------	----------------

R2	G0/2	192.168.6.1	192.168.6.0/24
----	------	-------------	----------------

R3	G0/0	192.168.3.3	192.168.3.0/24
----	------	-------------	----------------

R3	G0/1	192.168.6.2	192.168.6.0/24
----	------	-------------	----------------

R3	G0/2	192.168.7.1	192.168.7.0/24
----	------	-------------	----------------

Router Interface IP Address Connected Network

R4	G0/0	192.168.4.3	192.168.4.0/24
R4	G0/1	192.168.7.2	192.168.7.0/24

4. Static Routing Configuration

Static routes were configured manually on all four routers using the ip route command.

R1 Configuration

```
ip route 192.168.2.0 255.255.255.0 192.168.5.2
ip route 192.168.3.0 255.255.255.0 192.168.5.2
ip route 192.168.4.0 255.255.255.0 192.168.5.2
ip route 192.168.6.0 255.255.255.0 192.168.5.2
ip route 192.168.7.0 255.255.255.0 192.168.5.2
```

R2 Configuration

```
ip route 192.168.1.0 255.255.255.0 192.168.5.1
ip route 192.168.3.0 255.255.255.0 192.168.6.2
ip route 192.168.4.0 255.255.255.0 192.168.6.2
ip route 192.168.7.0 255.255.255.0 192.168.6.2
```

R3 Configuration

```
ip route 192.168.1.0 255.255.255.0 192.168.6.1
ip route 192.168.2.0 255.255.255.0 192.168.6.1
ip route 192.168.4.0 255.255.255.0 192.168.7.2
ip route 192.168.5.0 255.255.255.0 192.168.6.1
```

R4 Configuration

```
ip route 192.168.1.0 255.255.255.0 192.168.7.1
ip route 192.168.2.0 255.255.255.0 192.168.7.1
ip route 192.168.3.0 255.255.255.0 192.168.7.1
ip route 192.168.5.0 255.255.255.0 192.168.7.1
```

```
ip route 192.168.6.0 255.255.255.0 192.168.7.1
```

5. Verification

The routing table can be checked using:

```
show ip route
```

Static routes are displayed with the letter S.

Connectivity between different networks can be tested using the ping command.

For example, from PC1:

```
ping 192.168.4.2
```

A successful reply confirms that PC1 can communicate with PC8 through the configured static routes.

Similarly, from PC8:

```
ping 192.168.1.1
```

Successful replies confirm communication in the reverse direction.

Result

Static routing was successfully configured between the four routers, and communication between all the different networks was verified using the ping command.

Q2. Write the difference between Hub, Switch and Router in tabular format. Explain why Router is the most intelligent and Hub is the least intelligent device.

Answer:

Basis	Hub	Switch	Router
OSI Layer	Physical Layer (Layer 1)	Data Link Layer (Layer 2)	Network Layer (Layer 3)
Main Function	Connects multiple devices in a network	Connects devices and forwards data to the correct device	Connects different networks and forwards packets between them

Basis	Hub	Switch	Router
Address Used	Does not use MAC/IP address for forwarding	Uses MAC address	Uses IP address
Data Unit	Bits	Frames	Packets
Forwarding	Sends data to all connected ports	Sends data mainly to the destination port	Forwards packets through the best route
Collision	High possibility of collisions	Reduces collisions	Separates networks and collision domains
Broadcast Domain	Single broadcast domain	Single broadcast domain by default	Separates broadcast domains
Intelligence	Least intelligent	More intelligent than Hub	Most intelligent
Network Connection	Usually connects devices within a LAN	Connects devices within a LAN	Connects different networks
Example	Ethernet Hub	LAN Switch	Home/Internet Router

a) Why is Router the most intelligent device?

A **Router** is considered the most intelligent because it operates at the **Network Layer (Layer 3)** and uses **IP addresses** to make forwarding decisions.

It maintains a **routing table** and determines the appropriate or best path for transferring packets from one network to another. It can also connect different networks and separate broadcast domains.

b) Why is Hub the least intelligent device?

A **Hub** is the least intelligent because it operates only at the **Physical Layer (Layer 1)**. It does not understand MAC addresses or IP addresses.

When a hub receives data, it simply **broadcasts/repeats the signal to all connected ports**, without determining which device is the actual destination.

Q3. What are Random Access Protocols? Why do we need them? Explain each category in detail with diagram and example. Also explain the difference between CSMA/CD and CSMA/CA.

Answer:

Random Access Protocols are **Medium Access Control (MAC) protocols** used when multiple devices share a common communication channel.

In these protocols, any device can attempt to access the shared medium whenever it has data to transmit. Since multiple devices may try to transmit at the same time, **collisions may occur**.

The main purpose of Random-Access Protocols is to **control access to the shared medium and handle or reduce collisions**.

a) Why do we need Random Access Protocols?

When multiple devices share the same communication medium, two or more devices may try to transmit data at the same time. This can cause a **collision**, which may corrupt the transmitted data.

Therefore, Random Access Protocols are required to:

1. Control access to a shared communication medium.
2. Reduce or handle collisions.
3. Improve network efficiency.
4. Allow multiple devices to share the same communication channel.

b) Categories of Random Access Protocols

The major categories of Random Access Protocols are:

The main Random Access Protocols are:

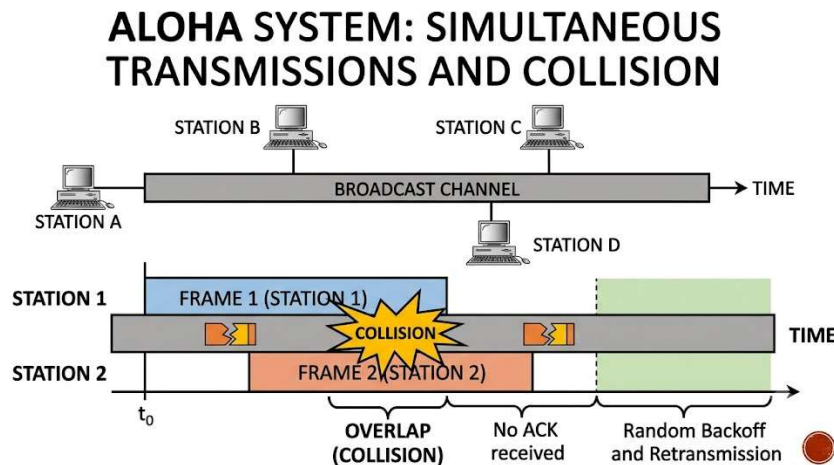
1. **ALOHA**
2. **CSMA**
3. **CSMA/CD**
4. **CSMA/CA**

1. ALOHA

ALOHA is one of the simplest Random Access Protocols.

In ALOHA, a device can transmit data whenever it has data to send. It does **not first check whether the channel is free**.

If two devices transmit at the same time, a **collision** occurs.



After a collision, the devices wait for a **random amount of time** and retransmit.

Working of ALOHA:

1. A station gets data to transmit.
2. It transmits the data immediately.
3. It waits for an acknowledgement.
4. If acknowledgement is received, transmission is successful.
5. If acknowledgement is not received, a collision is assumed.
6. The station waits for a random backoff time.
7. It retransmits the data.

Example:

Suppose PC1 and PC2 share the same communication channel.

- PC1 starts transmitting.
- PC2 also starts transmitting at the same time.
- Their signals overlap.

- A collision occurs.
- Both wait for a random amount of time.
- They retransmit their data.

Types of ALOHA

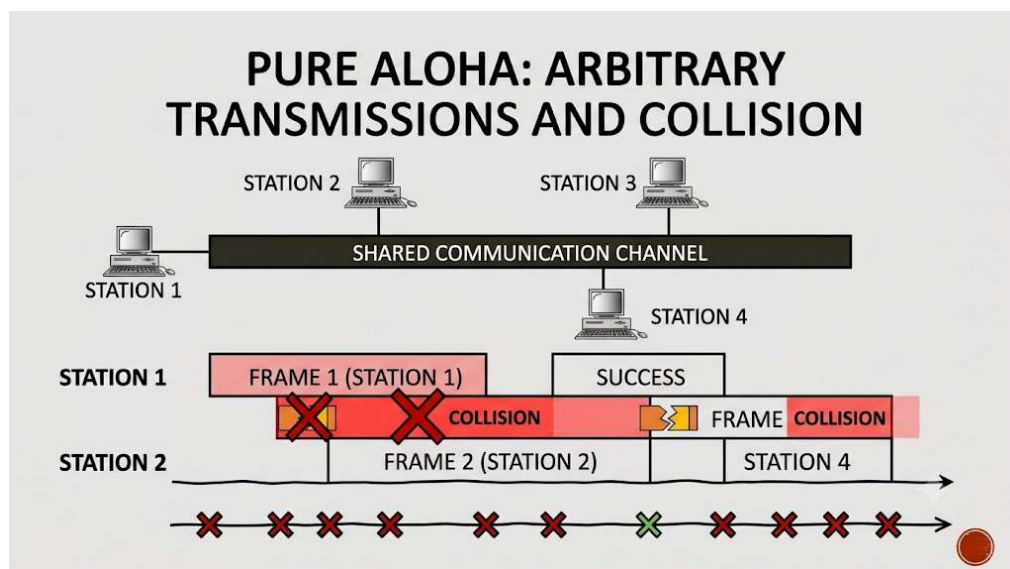
There are two types of ALOHA:

1. **Pure ALOHA**
2. **Slotted ALOHA**

a) Pure ALOHA

In **Pure ALOHA**, a device can transmit **at any time** when it has data to send.

There is no requirement to wait for a specific time slot.



Since devices can transmit at any time, the probability of collision is relatively high.

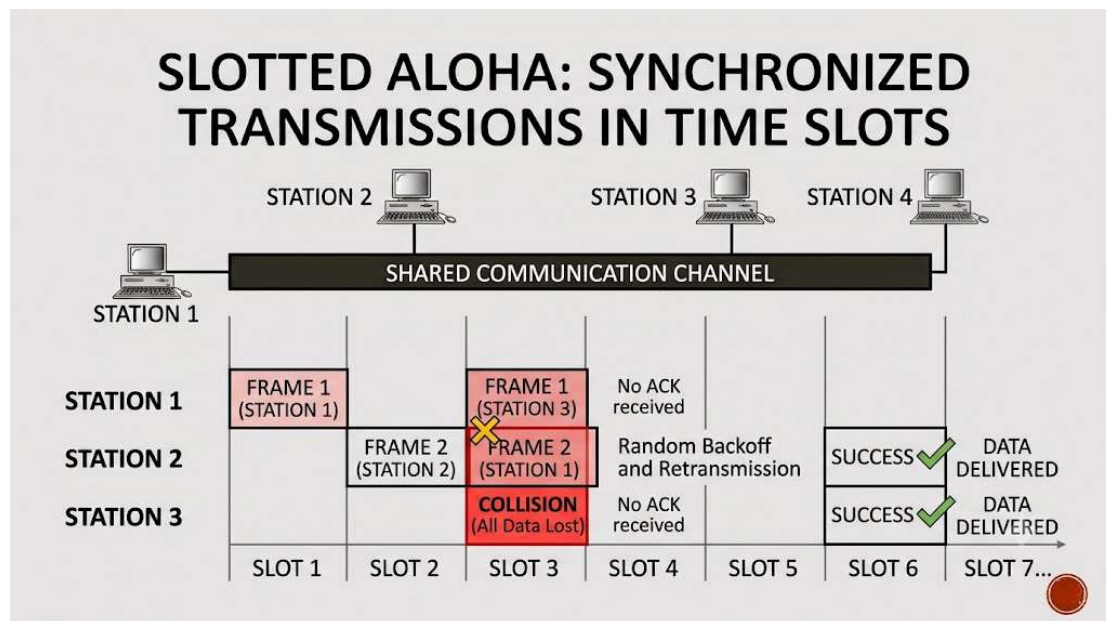
Example:

If PC1 starts transmitting at one point in time and PC2 starts transmitting shortly afterward, their frames may overlap, resulting in a collision.

b) Slotted ALOHA

In **Slotted ALOHA**, time is divided into equal-sized **time slots**.

A device can transmit only at the **beginning of a time slot**.



This reduces the possibility of collisions compared with Pure ALOHA.

Example:

If a station has data to send during a particular time slot, it waits until the beginning of the next appropriate slot before transmitting.

Difference between Pure ALOHA and Slotted ALOHA

Basis	Pure ALOHA	Slotted ALOHA
Transmission	Any time	Beginning of a time slot
Synchronization	Not required	Required
Collision Probability	Higher	Lower
Efficiency	Lower	Higher

2. CSMA

CSMA stands for **Carrier Sense Multiple Access**.

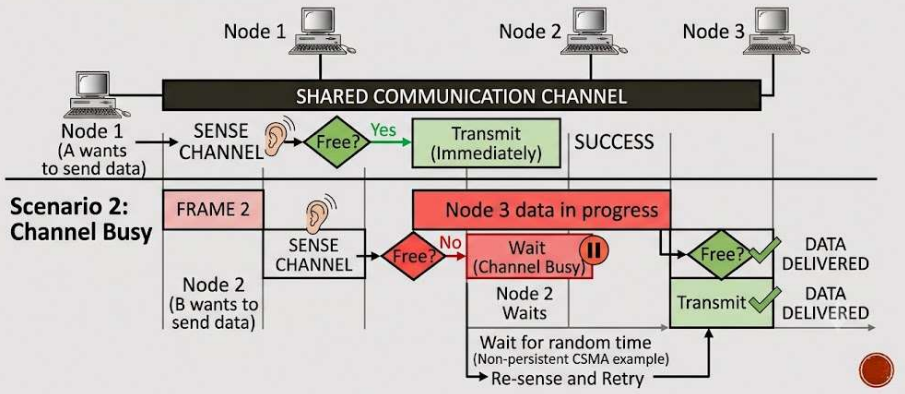
In CSMA, a device **senses or checks the channel before transmitting**.

The basic principle is:

"Listen before you transmit."

CSMA WORKING: SENSE, WAIT, OR TRANSMIT

Scenario 1: Channel Free



Working of CSMA:

1. A station has data to transmit.
2. It first senses the communication channel.
3. If the channel is free, it transmits.
4. If the channel is busy, it waits.
5. After the channel becomes free, the station can transmit.

Example:

If PC1 wants to send data, it first checks the channel.

- If no other device is transmitting → PC1 transmits.
- If another device is transmitting → PC1 waits.

Limitation of CSMA

CSMA cannot completely eliminate collisions.

Two devices may check the channel at almost the same time, both find it free, and then transmit simultaneously.

3. CSMA/CD

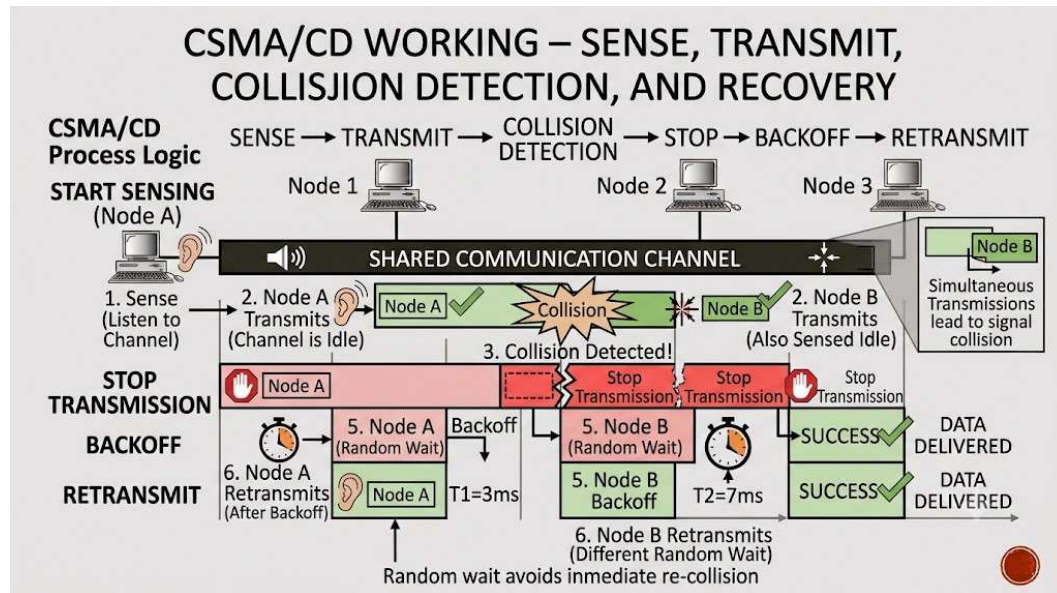
CSMA/CD stands for:

Carrier Sense Multiple Access with Collision Detection

It is traditionally associated with **shared, half-duplex Ethernet**.

The basic principle is:

"Listen before transmitting and detect collision while transmitting."



Working of CSMA/CD:

1. A station has data to transmit.
2. It senses the channel.
3. If the channel is busy, it waits.
4. If the channel is free, it starts transmitting.
5. While transmitting, it monitors the channel for a collision.
6. If no collision occurs, transmission is completed.
7. If a collision occurs, transmission is stopped.
8. The station waits for a random backoff period.
9. It retransmits the data.

Example:

Suppose PC1 and PC2 want to transmit:

- Both sense the channel.
- Both find it free.
- Both start transmitting.
- Their signals collide.

- The collision is detected.
- Both stop transmitting.
- They wait for a random backoff period.
- They retransmit.

4. CSMA/CA

CSMA/CA stands for:

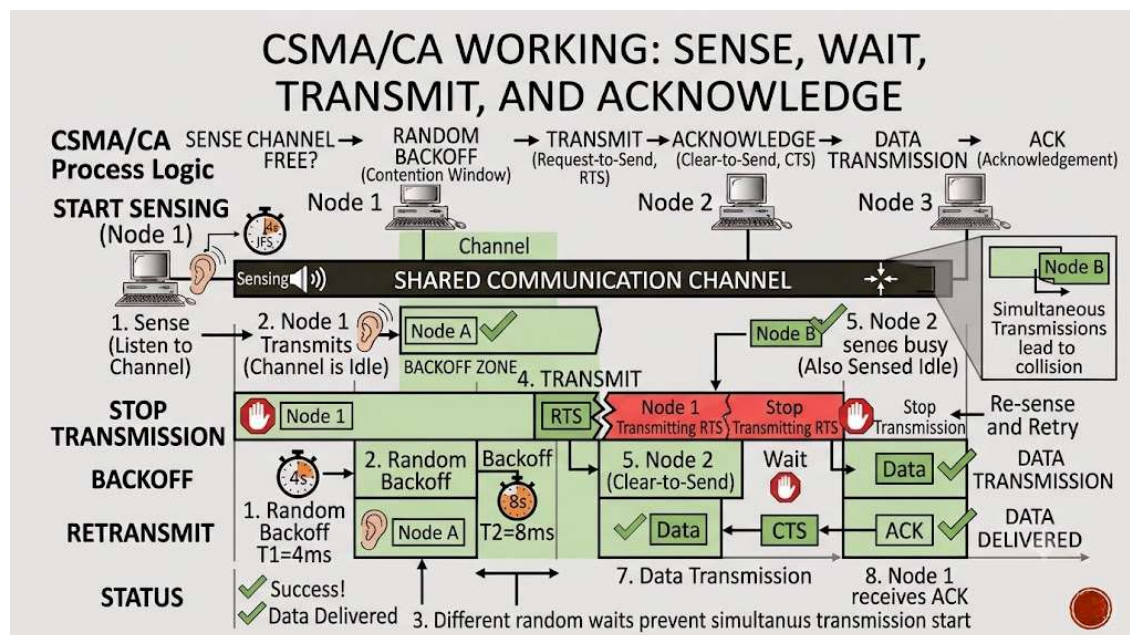
Carrier Sense Multiple Access with Collision Avoidance

It is commonly used in **Wireless LAN (Wi-Fi)**.

The basic principle is:

"Listen before transmitting and try to avoid collision."

Wireless devices generally cannot reliably detect collisions while transmitting, so CSMA/CA attempts to **avoid collisions before transmission**.



Working of CSMA/CA:

1. A station has data to transmit.
2. It senses the wireless channel.
3. If the channel is busy, it waits.
4. If the channel is free, it waits for a suitable backoff period.

5. It transmits the data.
6. The receiver sends an **ACK** after successful reception.
7. If the sender does not receive an ACK, it assumes that transmission was unsuccessful and tries again.

Example:

Suppose two laptops are connected to the same Wi-Fi network.

- Both laptops sense the wireless channel.
- If the channel is busy, they wait.
- If the channel becomes free, they use a random backoff period.
- The laptop whose backoff expires first transmits.
- The receiver sends an ACK after successful reception.

c) Difference Between CSMA/CD and CSMA/CA

Basis	CSMA/CD	CSMA/CA
Full Form	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Main Concept	Detects collisions	Attempts to avoid collisions
Commonly Used In	Traditional shared Ethernet	Wireless LAN / Wi-Fi
Collision Handling	Detects collision after it occurs	Attempts to prevent collision before it occurs
Method	Detect → Stop → Backoff → Retransmit	Sense → Wait/Backoff → Transmit → ACK
Medium	Wired shared medium	Wireless shared medium
ACK	Not the fundamental collision-detection mechanism	Commonly uses ACK to confirm successful reception
Reason	Collision detection is possible in traditional shared Ethernet	Collision detection is difficult while transmitting wirelessly